Annex III. Debt Sustainability Analysis—Update

This Debt Sustainability Analysis (DSA) update reflects revisions since the previous DSA completed in September 2014 for the 2014 Article IV consultation, to reflect updated information on the fiscal deficit and debt path, the release of revised national accounts data, and prospective borrowings under a precautionary Fund arrangement. Kenya's risk of external debt distress remains low, while overall public sector debt dynamics continue to be sustainable.[1] However, data gaps in the financial accounts of the balance of payments and the international investment position need to be addressed promptly to contain risks of an unmonitored buildup in nongovernment external debt.

UNDERLYING ASSUMPTIONS

1. This DSA is based on macroeconomic assumptions that are consistent with the framework outlined in the accompanying staff report. Main changes compared with the September 2014 DSA include (text table):

Kenya: Selected Macroeconomic Assumptions

	2012	2013	2014	2015	Long term 1/
Real GDP Growth					
Current DSA	4.5	5.7	5.3	6.9	6.8
Previous DSA (September 2014) 2/	4.5	4.6	5.0	5.8	7.0
Primary Fiscal Deficit (percent of GDP)					
Current DSA	2.3	3.0	4.5	5.1	0.7
Previous DSA (September 2014)	2.8	3.5	3.2	2.3	0.9
Non-interest Current Account Deficit (percent of GDP)					
Current DSA	8.2	8.4	8.3	6.8	5.6
Previous DSA (September 2014)	8.2	8.4	7.3	8.1	5.8

Source: IMF staff estimates.
1/ For current DSA, average 2020-34. For previous DSA, average 2019-33.
2/ Fiscal-year basis, prior to release of upwardly revised GDP statistics in September 2014.

- **Frontloading of public infrastructure spending.** Projected spending on several key infrastructure projects, including the Nairobi-Mombasa Standard Gauge Railway (SGR; see

[1] The World Bank in 2013 upgraded its classification of Kenya to "strong" in terms of the quality of its policies and institutions as measured by a three-year average of the World Bank's Country Policy and Institutional Assessment (CPIA) Index. The relevant indicative thresholds for this category are: 50 percent for the NPV of debt-to-GDP ratio, 200 percent for the NPV of debt-to-exports ratio, 300 percent for the NPV of debt-to-revenue ratio, 25 percent for the debt service-to-exports ratio, and 22 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt.

Box 2 and Annex II of the accompanying Staff Report), is more frontloaded than in the previous DSA. This results in a higher primary fiscal deficit in the near term, with the deficit being brought down in the medium to longer term consistent with the East African Community (EAC) Monetary Union convergence criteria.

- **Revised current account path.** The current account path reflects weaker than previously projected goods and services exports in 2014, higher imports related to frontloaded infrastructure spending, and lower imports fuel-related imports to reflect lower oil prices.

- **Revised national accounts.** This DSA incorporates the comprehensive revisions to the national accounts officially released at the end of September 2014. The previous DSA anticipated the upward revisions to GDP of some 20 percent, but with estimates only.

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

2. All indicators of public and publicly guaranteed external debt remain well below the policy-dependent debt burden thresholds under the baseline scenario, and no thresholds are breached under any of the standard stress tests. The main results of the external DSA are the following:

- **Under the baseline scenario, the debt burden remains sustainable over the 20-year projection period** (Figure 1). As a result of the issuances of the sovereign bond and railway-related loan package, the NPV of public and publicly guaranteed external debt is estimated at 18 percent of GDP at end-2014 and projected to peak at just over 20 percent in 2016–17 (well below the 50 percent indicative threshold). Nonresident holdings of domestic debt are reportedly low (Annex I). The NPV of the debt-to-exports ratio would plateau at just over 100 percent in the medium term, remaining well under an indicative threshold of 200 percent.

- **Standard stress tests do not reveal significant vulnerabilities as even the shocks with the highest impact would maintain debt levels below the relevant indicative thresholds** (Table 2 and Figure 1). The shocks that would have the largest impact on external debt dynamics result from a one-time 30 percent nominal depreciation of the exchange rate (similar to the previous DSA) and from a permanent shock to the terms of new public sector loans, increasing the PV of debt to GDP ratio from 18 percent in 2014 to nearly 30 percent (by 2016 for a depreciation shock, between 2023 and 2034 for worse terms), but still well below the relevant threshold (50 percent).

- **The current account balance could improve significantly if recent resource discoveries are confirmed as profitable.** Kenya is currently a net oil importer. However, an oil company has reported estimated discovered resources of at least 600 million barrels of oil equivalent, and oil and gas exploration activities are continuing. If recent discoveries are confirmed as commercially viable, Kenya could become self-sufficient in 3–5 years and a net exporter in 5–

10 years. This represents significant upside potential for the medium to long term compared with the baseline.

3. The picture is unclear for private sector debt. Available data from the Foreign Investment Survey suggested nongovernment external debt of some 10 percent of GDP as at end-2011. Stock data for later years are not yet available, nor or reliable flow data for the financial accounts of the balance of payments. However, robust growth in private sector activity and banks' balance sheets suggest that nongovernment external debt is likely on an upward trend. Data gaps should be addressed promptly to limit the risk of an unmonitored buildup of external vulnerabilities outside the government sector.

PUBLIC DEBT SUSTAINABILITY ANALYSIS

4. Public debt peaks in 2015 and subsequently remains on a downward trend in the baseline scenario. In 2014, overall public debt is estimated to have increased markedly to 50 percent of GDP, owing to the sovereign bond issuances[2] in June and December, and initial disbursements of the SGR-related loan from China's Eximbank. Subsequent disbursements are projected to lift public debt to just over 51 percent of GDP in 2015 (Table 3). In subsequent years, on the basis of a primary deficit brought back below 4 percent of GDP in 2016 and gradually brought down thereafter consistent with the EAC convergence criteria—as well as robust real GDP growth boosted by infrastructure spending and easing of bottlenecks—public debt eases below 50 percent of GDP by 2018. In PV terms, the public debt-to-GDP ratio would peak around 46 percent in 2015–16 (Table 4), easing thereafter. The PV of public debt-to-revenue ratio would gradually decline from around 220 percent in 2014–15 to around 190 percent in 2018.

5. The projected path of public debt is moderate in the Low-Income Country (LIC) DSA framework.

- The projected debt path would be below the EAC public debt convergence criterion (ceiling of 50 percent in PV terms), and is also well below the LIC DSA public debt benchmark for those countries whose CPIA score for quality of policies and institutions is assessed as strong (74 percent of GDP, also in PV terms) above which the risk of public debt distress is heightened. . However, gross financing needs are relatively high, at some 18 percent of GDP in 2014–15.

- The LIC DSA framework currently remains relevant for Kenya, since a majority of its external public debt remains concessional or semi-concessional. However, the 2014 sovereign bond

[2] The June 2014 issuance comprised two tranches: a five-year $500 million bond at a yield of 5.875 percent, and a 10-year $1.5 billion bond at 6.875 percent. In December 2014, Kenya added $250 million to the five-year tranche at a 5.0 percent yield and $500 million to the 10-year tranche at 5.9 percent.

issuances and the commercial component of the SGR have resulted in a significant rise in the share of Kenya's commercial external debt.

6. The alternative scenarios and bound tests indicate that the projected paths for all debt indicators remain under the relevant thresholds (Table 4 and Figure 2). The shock with the highest sustained negative impact on debt dynamics is real GDP growth one standard deviation below the historical average. This shock pushes the ratio of PV of debt-to-GDP to 52 percent from 2016 to 2019, above the EAC convergence criterion ceiling; the PV of debt-to-revenue ratio to 232 percent, and the PV of debt service-to-revenue ratio to nearly 30 percent. An unchanged primary balance from 2014 would have broadly similar results; indeed the authorities envisage a higher primary balance in 2015 owing to the SGR project, while planning to reduce the primary balance in the medium term consistent with the convergence criteria for the EAC monetary union.

7. Other fiscal challenges, reported in the previous DSA, remain pertinent:

- The devolution process has made concrete progress, but several issues—notably relating to wage bills, the county borrowing framework, and extension of the coverage of fiscal accounts to all county governments —need to be addressed to ensure the medium-term reduction in the primary balance remains within reach.

- Contingent liabilities could represent an additional fiscal risk. The extent of contingent liabilities stemming from Public Private Partnerships (PPPs), mostly in the energy sector, is yet to be fully assessed.

MAIN FINDINGS AND CONCLUSIONS

8. This DSA update finds that Kenya continues to face low risk of external debt distress. Institutional improvements in recent years have increased the safety margin between Kenya's projected debt and the related indicative thresholds, though the use of a higher discount rate also contributes significantly to the more favorable dynamics. Upward revisions to GDP time series have also lowered the ratios of debt to GDP.

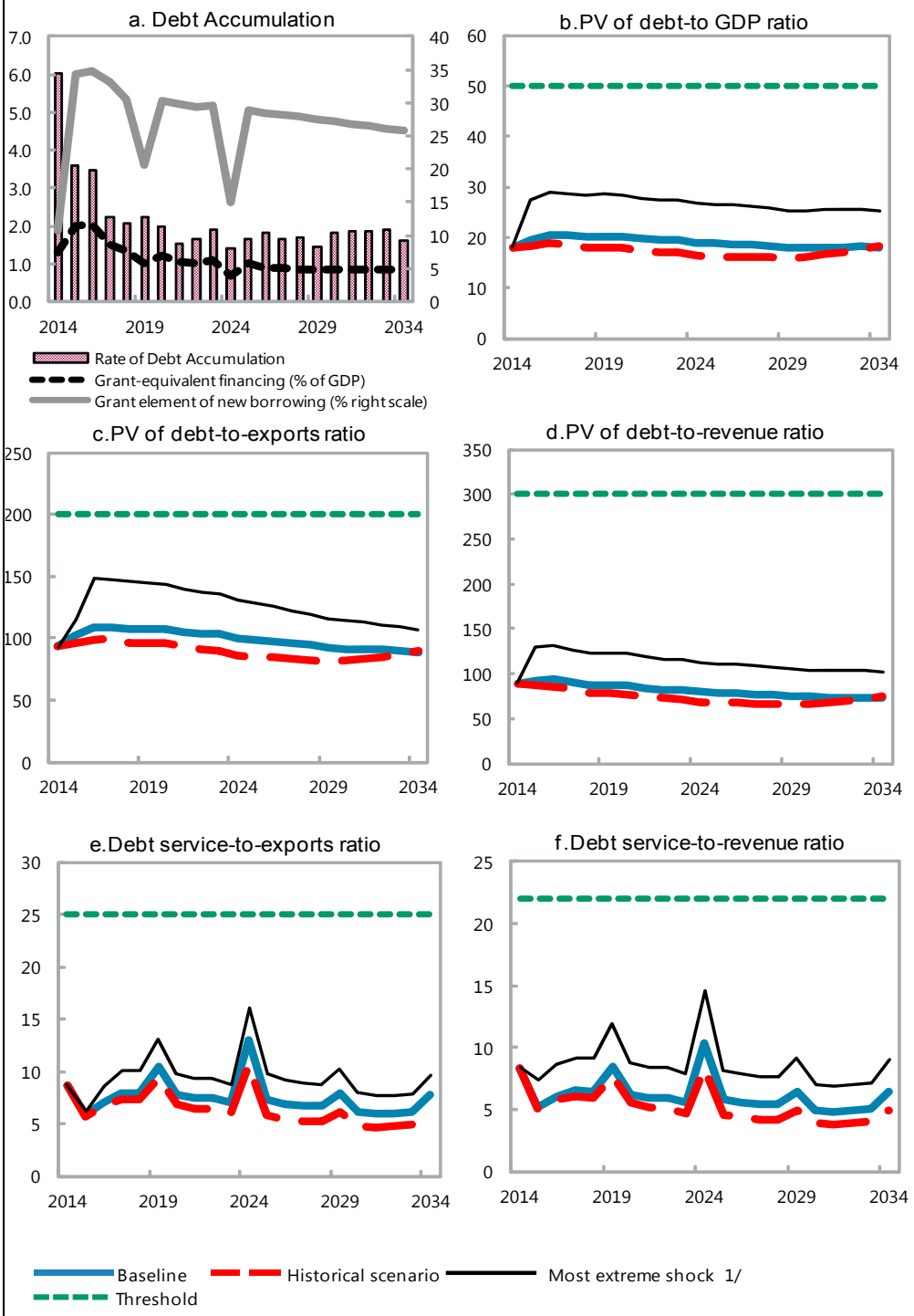
9. Standard stress tests suggest scenarios in which debt would increase, but remain within sustainable bounds. Exchange rate shocks and less favorable terms on new public sector loans represent the largest upside risks to external debt. Overall public debt would increase most in the event of significantly lower than anticipated economic growth. At the same time, Kenya has strong market foundations and long-standing sound macroeconomic policies—absence of price controls, flexible exchange rate and interest rates, limited budget subsidies—which give it scope to respond to shocks.

10. In the near to medium term, risks to debt dynamics are to the upside. In the near term, the fiscal deficit and borrowing needs could widen further if execution rates on foreign-financed projects (for which debt has been contracted but not disbursed) rise faster than expected; if management of the devolution process falters and the new county borrowing framework lacks

sufficient safeguards; and/or if risks materialize from contingent liabilities. In addition, the picture is particularly uncertain with regard to nongovernment external debt, for which data gaps should be addressed as soon as possible in order to limit risks of an unmonitored buildup in external vulnerabilities.

11. But natural resource discoveries represent positive potential for the medium to long term. If confirmed, resource exports could significantly improve Kenya's external prospects. This puts a premium on appropriate design for policies to manage Kenya's natural resources.

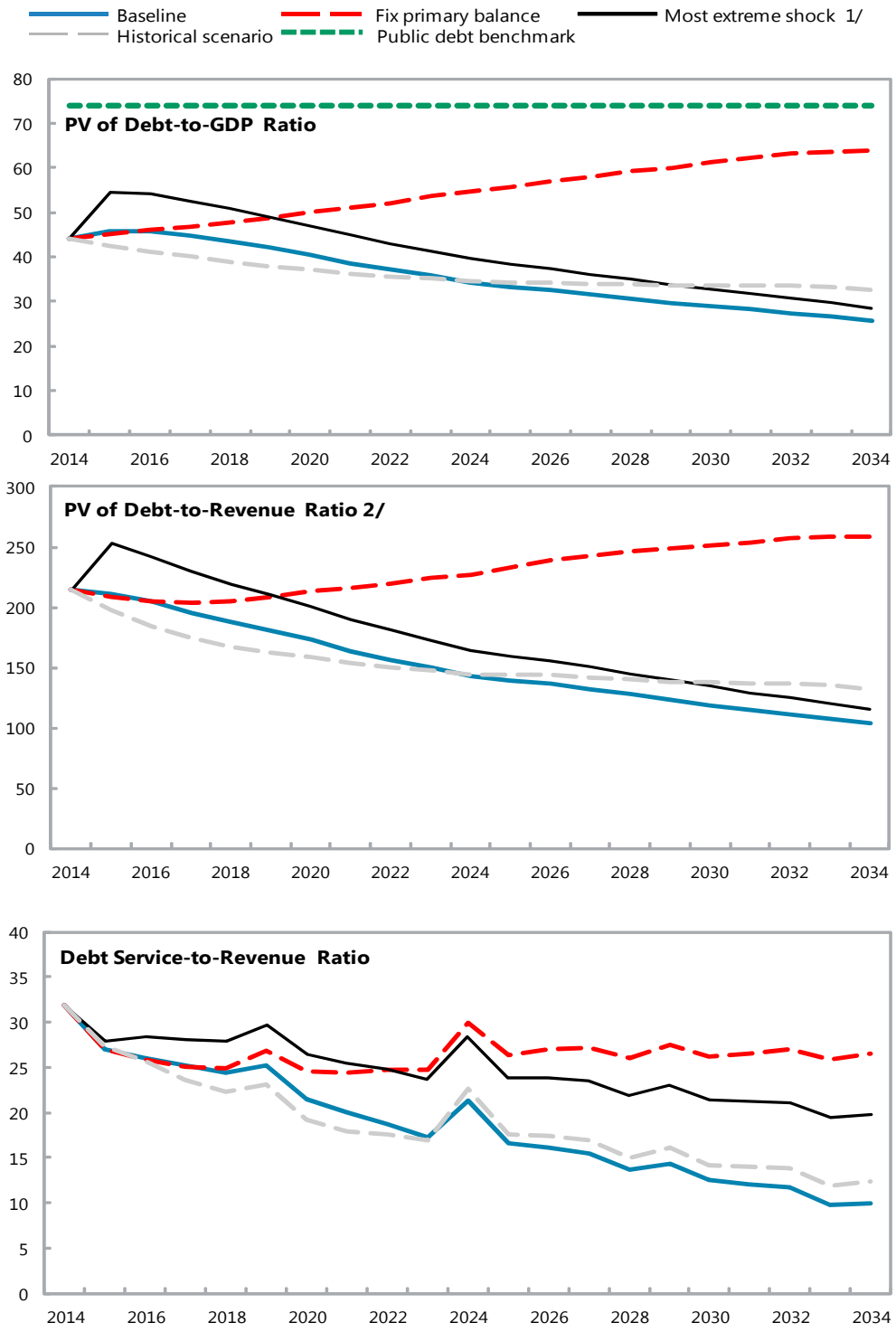
Figure 1. Kenya: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2014-2034--Baseline 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Kenya: Indicators of Public Debt Under Alternative Scenarios, 2014-2034--Baseline 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 1. Kenya: External Debt Sustainability Framework, Baseline Scenario, 2011-2034 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**27.1**	**31.6**	**32.4**			**37.6**	**41.5**	**42.1**	**42.4**	**42.7**	**43.0**		**43.1**	**44.2**	
of which: public and publicly guaranteed (PPG)	18.4	19.3	19.4			23.8	25.1	26.4	26.4	26.5	26.5		25.4	24.0	
Change in external debt	-0.2	4.5	0.8			5.2	3.9	0.6	0.3	0.3	0.3		-0.6	0.0	
Identified net debt-creating flows	7.1	3.4	4.9			6.0	3.3	2.6	2.0	2.7	2.0		2.2	1.1	
Non-interest current account deficit	**8.9**	**8.2**	**8.4**	**4.6**	**3.2**	**8.3**	**6.8**	**6.5**	**5.9**	**6.8**	**6.0**		**6.0**	**4.4**	5.6
Deficit in balance of goods and services	15.4	13.7	13.7			13.3	11.9	11.8	11.7	12.3	11.4		11.0	8.8	
Exports	23.6	21.9	19.4			19.3	19.1	19.0	18.7	18.7	18.8		18.9	20.4	
Imports	39.0	35.6	33.2			32.6	31.0	30.8	30.4	31.0	30.1		29.9	29.1	
Net current transfers (negative = inflow)	-6.4	-5.6	-5.7	-6.1	0.5	-5.3	-5.3	-5.2	-5.1	-4.9	-4.8		-4.4	-3.6	-4.1
of which: official	-0.4	-0.4	-0.4			-0.1	-0.1	-0.1	0.0	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	-0.1	0.1	0.3			0.4	0.1	-0.2	-0.7	-0.6	-0.6		-0.6	-0.7	
Net FDI (negative = inflow)	**-0.8**	**-0.5**	**-0.9**	**-0.6**	**0.6**	**-1.1**	**-1.7**	**-1.9**	**-2.0**	**-2.1**	**-2.1**		**-1.8**	**-1.1**	-1.6
Endogenous debt dynamics 2/	**-1.0**	**-4.3**	**-2.5**			**-1.3**	**-1.9**	**-2.0**	**-2.0**	**-2.0**	**-2.0**		**-2.0**	**-2.3**	
Contribution from nominal interest rate	0.2	0.3	0.3			0.3	0.5	0.6	0.7	0.7	0.7		0.7	0.6	
Contribution from real GDP growth	-1.6	-1.0	-1.7			-1.5	-2.3	-2.7	-2.7	-2.7	-2.7		-2.7	-2.8	
Contribution from price and exchange rate changes	0.3	-3.5	-1.2			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-7.3**	**1.0**	**-4.2**			**-0.8**	**0.7**	**-2.0**	**-1.7**	**-2.4**	**-1.7**		**-2.8**	**-1.1**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.1	0.0	
PV of external debt 4/	...	...	26.5			31.8	35.9	36.2	36.3	36.4	36.7		36.7	38.1	
In percent of exports	...	...	136.3			164.2	187.9	190.8	193.6	194.4	195.4		193.6	187.3	
PV of PPG external debt	**...**	**...**	**13.5**			**18.0**	**19.4**	**20.5**	**20.3**	**20.2**	**20.2**		**18.9**	**17.9**	
In percent of exports	**...**	**...**	**69.5**			**93.1**	**101.7**	**108.0**	**108.5**	**107.8**	**107.7**		**100.1**	**88.1**	
In percent of government revenues	**...**	**...**	**70.5**			**89.5**	**92.0**	**93.3**	**89.9**	**87.7**	**87.4**		**79.4**	**72.9**	
Debt service-to-exports ratio (in percent)	**11.2**	**14.1**	**18.1**			**24.8**	**25.5**	**25.6**	**26.5**	**26.8**	**29.4**		**30.4**	**26.1**	
PPG debt service-to-exports ratio (in percent)	**3.5**	**3.7**	**4.0**			**8.7**	**5.8**	**7.1**	**7.9**	**8.0**	**10.5**		**13.0**	**7.8**	
PPG debt service-to-revenue ratio (in percent)	**4.4**	**4.3**	**4.0**			**8.4**	**5.2**	**6.1**	**6.5**	**6.5**	**8.5**		**10.3**	**6.4**	
Total gross financing need (Billions of U.S. dollars)	5.8	7.0	8.7			10.4	10.2	11.5	12.3	14.4	15.8		27.0	67.5	
Non-interest current account deficit that stabilizes debt ratio	9.1	3.7	7.6			3.1	2.9	6.0	5.6	6.5	5.7		6.6	4.5	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.1	4.5	5.7	5.1	2.2	5.3	6.9	7.2	7.1	7.0	7.0	6.7	6.8	7.0	6.8
GDP deflator in US dollar terms (change in percent)	-1.2	14.9	3.8	7.4	7.1	4.7	2.9	4.3	4.3	3.9	3.4	3.9	3.0	3.0	3.0
Effective interest rate (percent) 5/	1.0	1.1	1.0	1.4	0.5	0.9	1.4	1.7	1.8	1.8	1.8	1.6	1.7	1.4	1.6
Growth of exports of G&S (US dollar terms, in percent)	10.3	11.3	-2.7	12.0	11.1	9.7	8.6	11.1	10.3	10.9	11.0	10.3	10.4	11.2	10.6
Growth of imports of G&S (US dollar terms, in percent)	20.8	9.6	2.2	16.3	12.0	8.5	4.5	11.0	10.2	13.2	7.6	9.2	9.9	9.8	9.8
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	10.7	34.2	34.9	33.1	30.6	20.6	27.4	15.0	25.7	27.2
Government revenues (excluding grants, in percent of GDP)	19.0	18.7	19.1			20.1	21.1	22.0	22.6	23.0	23.1		23.9	24.6	24.0
Aid flows (in Billions of US dollars) 7/	0.2	0.2	0.3			2.2	2.3	2.6	2.1	2.0	1.7		2.1	7.5	
of which: Grants	0.2	0.2	0.3			0.3	0.3	0.3	0.3	0.2	0.2		0.3	0.4	
of which: Concessional loans	0.0	0.0	0.0			1.8	2.0	2.3	1.8	1.7	1.5		1.8	7.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.3	2.0	2.0	1.5	1.3	1.0		0.7	0.9	0.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			16.8	40.7	40.0	38.4	35.3	24.6		19.4	27.8	30.0
Memorandum items:															
Nominal GDP (Billions of US dollars)	42.0	50.3	55.2			60.9	66.9	74.8	83.6	92.9	102.7		165.4	431.2	
Nominal dollar GDP growth	4.9	20.0	9.7			10.2	10.0	11.8	11.6	11.1	10.6	10.9	10.0	10.2	10.0
PV of PPG external debt (in Billions of US dollars)			7.4			10.8	13.0	15.3	16.9	18.7	20.7		31.3	77.2	
(PVt-PVt-1)/GDPt-1 (in percent)						6.0	3.6	3.5	2.2	2.1	2.2	3.3	1.4	1.6	1.7
Gross workers' remittances (Billions of US dollars)	0.9	1.2	1.3			1.4	1.6	1.8	2.0	2.2	2.3		3.5	7.8	
PV of PPG external debt (in percent of GDP + remittances)	...	...	13.2			17.6	19.0	20.0	19.9	19.7	19.8		18.6	17.6	
PV of PPG external debt (in percent of exports + remittances)	...	...	62.0			83.0	90.5	96.0	96.3	95.8	96.1		90.0	80.9	
Debt service of PPG external debt (in percent of exports + remittanc	...	...	3.6			7.8	5.1	6.3	7.0	7.1	9.3		11.7	7.1	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Kenya: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Baseline Scenario 2014-2034
(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	18	19	20	20	20	20	**19**	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	18	18	19	19	18	18	**16**	18
A2. New public sector loans on less favorable terms in 2014-2034 2	18	20	22	22	23	24	**26**	28
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	18	20	22	22	22	22	**20**	19
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	18	20	24	23	23	23	**21**	18
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	18	20	22	22	21	22	**20**	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	18	20	23	22	22	22	**20**	18
B5. Combination of B1-B4 using one-half standard deviation shocks	18	21	25	24	24	24	**22**	19
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	18	27	29	29	28	28	**27**	25
PV of debt-to-exports ratio								
Baseline	93	102	108	108	108	108	**100**	88
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	93	95	99	100	96	96	**86**	90
A2. New public sector loans on less favorable terms in 2014-2034 2	93	104	114	119	123	127	**135**	139
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	93	101	108	108	107	107	**100**	88
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	93	115	148	148	146	145	**131**	107
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	93	101	108	108	107	107	**100**	88
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	93	107	119	119	118	117	**107**	90
B5. Combination of B1-B4 using one-half standard deviation shocks	93	109	131	130	129	128	**116**	96
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	93	101	108	108	107	107	**100**	88
PV of debt-to-revenue ratio								
Baseline	89	92	93	90	88	87	**79**	73
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	89	86	86	83	78	78	**69**	74
A2. New public sector loans on less favorable terms in 2014-2034 2	89	94	98	98	100	103	**107**	115
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	89	95	101	97	95	94	**86**	79
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	89	97	108	103	100	99	**88**	75
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	89	94	99	96	93	93	**85**	78
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	89	97	103	99	96	95	**85**	74
B5. Combination of B1-B4 using one-half standard deviation shocks	89	99	112	107	104	103	**91**	79
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	89	130	131	127	123	123	**112**	103

Table 2. Kenya: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Baseline Scenario, 2014-20134 (concluded)
Baseline Scenario, 2014-20134 (concluded)
(In percent)

Debt service-to-exports ratio								
Baseline	9	6	7	8	8	10	**13**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	9	6	7	7	7	9	**11**	6
A2. New public sector loans on less favorable terms in 2014-2034 2	9	6	6	7	7	7	**7**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	9	6	7	8	8	10	**13**	8
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	9	6	9	10	10	13	**16**	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	9	6	7	8	8	10	**13**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	9	6	7	8	8	11	**13**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	9	6	8	9	9	12	**14**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	9	6	7	8	8	10	**13**	8
Debt service-to-revenue ratio								
Baseline	8	5	6	7	6	8	**10**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	8	5	6	6	6	8	**8**	5
A2. New public sector loans on less favorable terms in 2014-2034 2	8	5	5	6	6	6	**6**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	8	5	7	7	7	9	**11**	7
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	8	5	6	7	7	9	**11**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	8	5	7	7	7	9	**11**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	8	5	6	7	7	9	**11**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	8	5	7	7	7	9	**11**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	8	7	9	9	9	12	**15**	9
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	22	22	22	22	22	22	**22**	22

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Kenya: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011-2034

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections									
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	40.2	41.8	43.9			50.0	51.3	51.7	50.9	49.8	48.4		40.7	31.7	
of which: foreign-currency denominated	18.4	19.3	19.4			23.8	25.1	26.4	26.4	26.5	26.5		25.4	24.0	
Change in public sector debt	-1.2	1.6	2.2			6.1	1.2	0.5	-0.8	-1.1	-1.4		-1.8	-1.0	
Identified debt-creating flows	-1.4	-0.1	1.3			2.7	2.0	0.4	-1.1	-1.2	-1.4		-1.8	-1.4	
Primary deficit	1.7	2.3	3.0	0.9	1.7	4.5	5.1	3.8	2.3	1.8	1.4	3.1	0.6	0.9	0.7
Revenue and grants	19.5	19.2	19.6			20.7	21.6	22.4	22.9	23.2	23.4		24.0	24.7	
of which: grants	0.5	0.4	0.5			0.5	0.5	0.4	0.3	0.3	0.2		0.2	0.1	
Primary (noninterest) expenditure	21.1	21.4	22.7			25.1	26.7	26.1	25.2	25.1	24.7		24.6	25.6	
Automatic debt dynamics	-3.1	-2.3	-1.7			-1.7	-3.1	-3.4	-3.4	-3.0	-2.8		-2.4	-2.3	
Contribution from interest rate/growth differential	-2.6	-1.3	-1.0			-1.5	-2.5	-2.8	-2.9	-2.6	-2.5		-2.0	-1.9	
of which: contribution from average real interest rate	-0.2	0.4	1.3			0.7	0.7	0.6	0.5	0.7	0.8		0.7	0.2	
of which: contribution from real GDP growth	-2.4	-1.7	-2.3			-2.2	-3.2	-3.4	-3.4	-3.3	-3.2		-2.7	-2.1	
Contribution from real exchange rate depreciation	-0.5	-1.0	-0.7			-0.3	-0.6	-0.6	-0.5	-0.4	-0.3		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.2	1.6	0.8			3.4	-0.7	0.1	0.3	0.1	0.0		0.0	0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	38.0			44.2	45.6	45.9	44.8	43.5	42.1		34.3	25.6	
of which: foreign-currency denominated	...	...	13.5			18.0	19.4	20.5	20.3	20.2	20.2		18.9	17.9	
of which: external	...	...	13.5			18.0	19.4	20.5	20.3	20.2	20.2		18.9	17.9	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	12.5	13.3	14.7			17.6	17.8	16.5	14.7	14.0	13.5		10.0	5.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	193.6			214.2	211.4	205.1	195.6	187.3	180.3		142.6	103.7	
PV of public sector debt-to-revenue ratio (in percent)	...	...	198.7			219.9	216.2	208.8	198.3	189.4	181.9		143.7	104.1	
of which: external 3/	...	...	70.5			89.5	92.0	93.3	89.9	87.7	87.4		79.4	72.9	
Debt service-to-revenue and grants ratio (in percent) 4/	25.9	27.7	28.7			31.8	27.0	26.0	25.2	24.4	25.2		21.3	9.9	
Debt service-to-revenue ratio (in percent) 4/	26.5	28.4	29.4			32.7	27.6	26.5	25.5	24.6	25.4		21.5	9.9	
Primary deficit that stabilizes the debt-to-GDP ratio	2.9	0.7	0.9			-1.7	3.9	3.3	3.1	2.9	2.8		2.4	1.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.1	4.5	5.7	5.1	2.2	5.3	6.9	7.2	7.1	7.0	7.0	6.7	6.8	7.0	6.8
Average nominal interest rate on forex debt (in percent)	1.2	1.5	1.4	1.5	0.5	1.2	2.0	2.5	2.6	2.8	2.8	2.3	2.8	2.6	2.7
Average real interest rate on domestic debt (in percent)	-0.1	2.3	5.9	1.5	3.7	3.5	2.4	1.9	1.8	2.4	2.7	2.4	2.5	-0.3	3.2
Real exchange rate depreciation (in percent, + indicates depreciation	-2.9	-5.9	-3.7	-4.5	7.6	-1.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	10.8	9.3	5.7	8.6	3.8	6.7	6.6	6.5	6.4	6.0	5.5	6.3	5.1	5.1	5.1
Growth of real primary spending (deflated by GDP deflator, in percer	1.9	5.9	11.9	2.0	3.9	16.5	13.6	4.8	3.4	6.4	5.4	8.4	7.5	8.0	7.1
Grant element of new external borrowing (in percent)	...	...	...	...	...	10.7	34.2	34.9	33.1	30.6	20.6	27.4	15.0	25.7	...

Sources: Country authorities; and staff estimates and projections.

1/ Refers to gross debt of the central government.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Kenya: Sensitivity Analysis for Key Indicators of Public Debt 2014-2034

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	44	46	46	45	44	42	34	26
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	44	43	41	40	39	38	35	32
A2. Primary balance is unchanged from 2014	44	45	46	47	48	49	55	64
A3. Permanently lower GDP growth 1/	44	46	47	46	45	44	40	44
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	44	48	52	52	52	52	50	49
B2. Primary balance is at historical average minus one standard deviations in 2015-201	44	43	43	42	41	40	32	25
B3. Combination of B1-B2 using one half standard deviation shocks	44	44	44	44	44	44	41	40
B4. One-time 30 percent real depreciation in 2015	44	53	52	50	49	47	38	29
B5. 10 percent of GDP increase in other debt-creating flows in 2015	44	54	54	53	51	49	40	29
PV of Debt-to-Revenue Ratio 2/								
Baseline	214	211	205	196	187	180	143	104
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	214	197	185	175	167	162	144	131
A2. Primary balance is unchanged from 2014	214	209	205	204	205	209	227	258
A3. Permanently lower GDP growth 1/	214	213	208	200	194	190	168	177
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	214	223	232	228	225	223	206	197
B2. Primary balance is at historical average minus one standard deviations in 2015-201	214	201	191	183	176	169	134	99
B3. Combination of B1-B2 using one half standard deviation shocks	214	203	195	192	189	187	171	161
B4. One-time 30 percent real depreciation in 2015	214	245	233	220	209	200	158	119
B5. 10 percent of GDP increase in other debt-creating flows in 2015	214	252	242	230	219	210	164	116
Debt Service-to-Revenue Ratio 2/								
Baseline	32	27	26	25	24	25	21	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	32	27	26	24	22	23	23	12
A2. Primary balance is unchanged from 2014	32	27	26	25	25	27	30	27
A3. Permanently lower GDP growth 1/	32	27	26	26	25	26	24	17
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	32	28	28	28	28	30	28	20
B2. Primary balance is at historical average minus one standard deviations in 2015-201	32	27	25	24	23	24	21	9
B3. Combination of B1-B2 using one half standard deviation shocks	32	28	27	24	23	25	25	16
B4. One-time 30 percent real depreciation in 2015	32	28	29	28	28	30	28	16
B5. 10 percent of GDP increase in other debt-creating flows in 2015	32	27	29	30	29	29	23	11

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.